

Monitoring of mercury, arsenic, and lead in traditional Asian herbal preparations on the Dutch market and estimation of associated risks.



Traditional herbal preparations used in Ayurveda, traditional Chinese medicine, traditional Tibetan medicine, and other Asian traditional medicine systems may contain significant amounts of mercury, arsenic or lead. Though deliberately incorporated in Asian traditional herbal preparations for therapeutic purposes, these constituents have caused intoxications worldwide. The aim of this study was therefore to determine mercury, arsenic, and lead levels in Asian traditional herbal preparations on the Dutch market. A total of 292 traditional herbal preparations used in Ayurveda, traditional Chinese medicine, and traditional Tibetan medicine were sampled between 2004 and 2007. Samples were mostly multi-ingredient traditional herbal preparations containing herbs and minerals. The labeling of less than 20% of the traditional herbal preparations suggested the presence of mercury, arsenic or lead. These elements were shown by inductively coupled mass spectrometry (ICP-MS) in 186 (64%) of 292 traditional herbal preparations. Estimated weekly mercury, arsenic, and lead intake levels were calculated for each traditional herbal preparation from the analytically determined concentrations and the recommended dose. A total of 59 traditional herbal preparations (20%) were likely to result in intakes of these elements significantly exceeding safety limits. Of these 59 traditional herbal preparations, intake estimates for 50 traditional herbal preparations significantly exceeded the safety limit for mercury (range = 1.4-1747 mg week⁽⁻¹⁾); intake estimates for 26 traditional herbal preparations significantly exceeded the safety limit for arsenic (range = 0.53-427 mg week⁽⁻¹⁾) and intake estimates for eight traditional herbal preparations were significantly above the safety limit for lead (range = 2.6-192 mg week⁽⁻¹⁾). It is concluded that the mercury, arsenic, and lead contents of traditional herbal preparations used in Ayurveda, traditional Chinese medicine, and traditional Tibetan medicine remain a cause for concern and require strict control.